

Exercise 05

Simulation and Analysis of Wireless Network using NS2 Simulator

Aim:

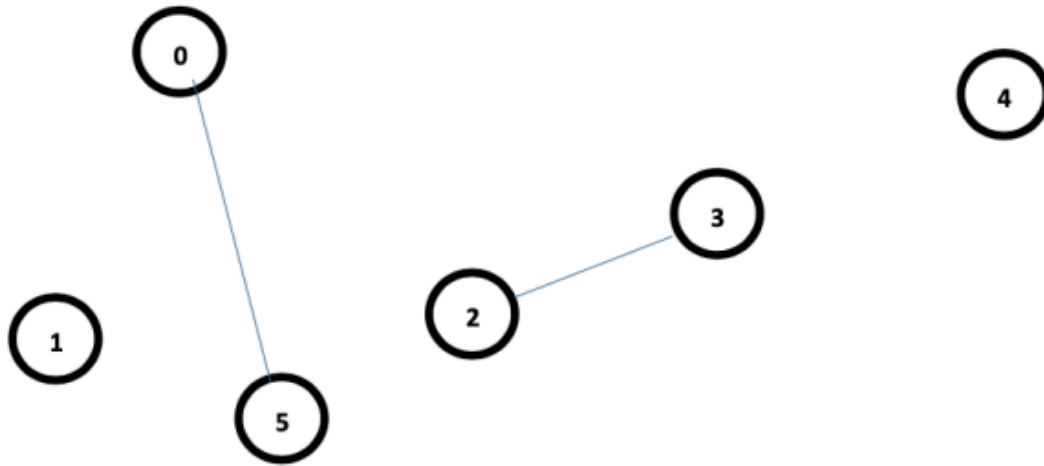
To simulate and analysis of wireless network using an NS2 simulator.

Procedure:

1. #Example of Wireless networks
2. initialize variables
3. Create a Simulator object
4. Create Tracing and animation file
5. topography
6. GOD - General Operations Director
7. Create nodes
8. Create Channel (Communication PATH)
9. Position of the nodes (Wireless nodes needs a location)
10. Any mobility codes (if the nodes are moving)
11. TCP, UDP Traffic

Flowchart:

Network Design:



Node 0, 2, 3 - Stationary Node

Node 1, 4, 5 - Mobile Node

Design Procedure:

Node 2 – Act as an agent and can send the UDP packet at CBR mode.

Node 0 – Act as an agent and can send TCP packet at FTP mode.

Node 3 – Receives the packet from Node 2. It is going to receive UDP packet hence it is NULL agent.

Node 5 – Receives the packet from Node 0. It is going to receive TCP packet hence it acts as Sink Agent.

Code:

Wireless_simulation.tcl

```
# Step 1: Initialize variables
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
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set val(ll) LL
set val(ant) Antenna/OmniAntenna

set val(ifqlen) 50
set val(nn) 6
set val(rp) AODV

set val(x) 500
set val(y) 500

# Step 2: Create Simulator
set ns [new Simulator]
# Step 3: Create Trace and NAM files
set tracefile [open wireless.tr w]
$ns trace-all $tracefile

set namfile [open wireless.nam w]
$ns namtrace-all-wireless $namfile $val(x) $val(y)
# Step 4: Create Topography
set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
# Step 5: Create GOD
create-god $val(nn)
# Step 6: Create Wireless Channel
set channel1 [new $val(chan)]
# Step 7: Configure Wireless Nodes
$ns node-config \
    -adhocRouting $val(rp) \
    -llType $val(ll) \
    -macType $val(mac) \
    -ifqType $val(ifq) \
    -ifqLen $val(ifqlen) \
```

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        -antType $val(ant) \
        -propType $val(prop) \
        -phyType $val(netif) \
        -topoInstance $topo \
        -agentTrace ON \
        -macTrace ON \
        -routerTrace ON \
        -movementTrace ON \
        -channel $channell
# Step 8: Create Nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
# Disable random movement
$n0 random-motion 0
$n1 random-motion 0
$n2 random-motion 0
$n3 random-motion 0
$n4 random-motion 0
$n5 random-motion 0
# Initial node positions
$ns initial_node_pos $n0 20
$ns initial_node_pos $n1 20
$ns initial_node_pos $n2 20
$ns initial_node_pos $n3 20
$ns initial_node_pos $n4 20
$ns initial_node_pos $n5 20
# Step 9: Set Initial Coordinates
$n0 set X_ 10.0

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\$n0 set Y_ 20.0

\$n0 set Z_ 0.0

\$n1 set X_ 210.0

\$n1 set Y_ 230.0

\$n1 set Z_ 0.0

\$n2 set X_ 100.0

\$n2 set Y_ 200.0

\$n2 set Z_ 0.0

\$n3 set X_ 150.0

\$n3 set Y_ 230.0

\$n3 set Z_ 0.0

\$n4 set X_ 430.0

\$n4 set Y_ 320.0

\$n4 set Z_ 0.0

\$n5 set X_ 270.0

\$n5 set Y_ 120.0

\$n5 set Z_ 0.0

Step 10: Mobility

At time 1 second:

n1 moves to (490,340) with speed 25 m/s

\$ns at 1.0 "\$n1 setdest 490.0 340.0 25.0"

n4 moves to (300,130) with speed 5 m/s

\$ns at 1.0 "\$n4 setdest 300.0 130.0 5.0"

```
# n5 moves to (190,440) with speed 15 m/s

$ns at 1.0 "$n5 setdest 190.0 440.0 15.0"

# At time 20 seconds:
# n5 moves to (100,200) with speed 30 m/s

$ns at 20.0 "$n5 setdest 100.0 200.0 30.0"
# Step 11: TCP + FTP Traffic
set tcp [new Agent/TCP]
set sink [new Agent/TCPSink]

$ns attach-agent $n0 $tcp
$ns attach-agent $n5 $sink

$ns connect $tcp $sink

set ftp [new Application/FTP]
$ftp attach-agent $tcp

$ns at 1.0 "$ftp start"
# Step 12: UDP + CBR Traffic
set udp [new Agent/UDP]
set null [new Agent/Null]

$ns attach-agent $n2 $udp
$ns attach-agent $n3 $null

$ns connect $udp $null

set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
```

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$ns at 1.0 "$cbr start"
# Step 13: Finish
$ns at 30.0 "finish"
proc finish {} {
    global ns tracefile namfile

    $ns flush-trace

    close $tracefile
    close $namfile

    exit 0
}
# Step 14: Run Simulation
puts "Starting Simulation"

$ns run

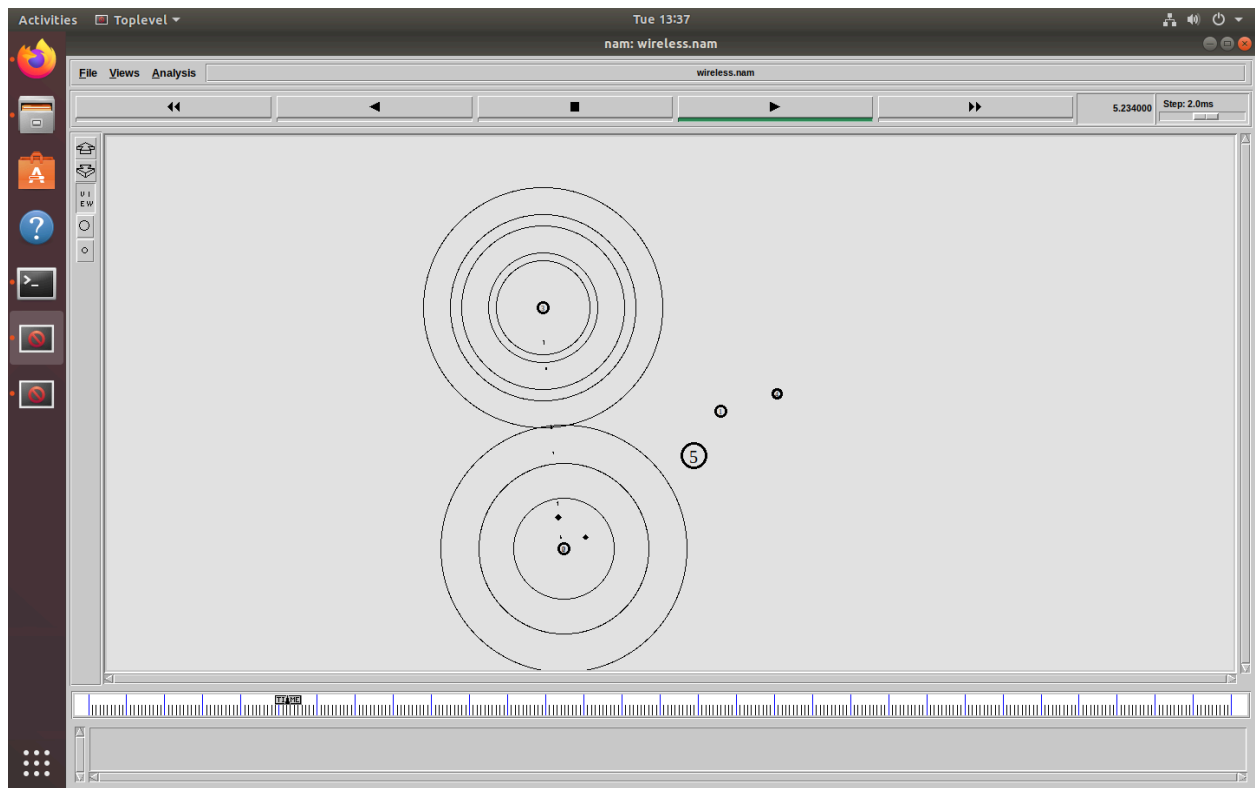
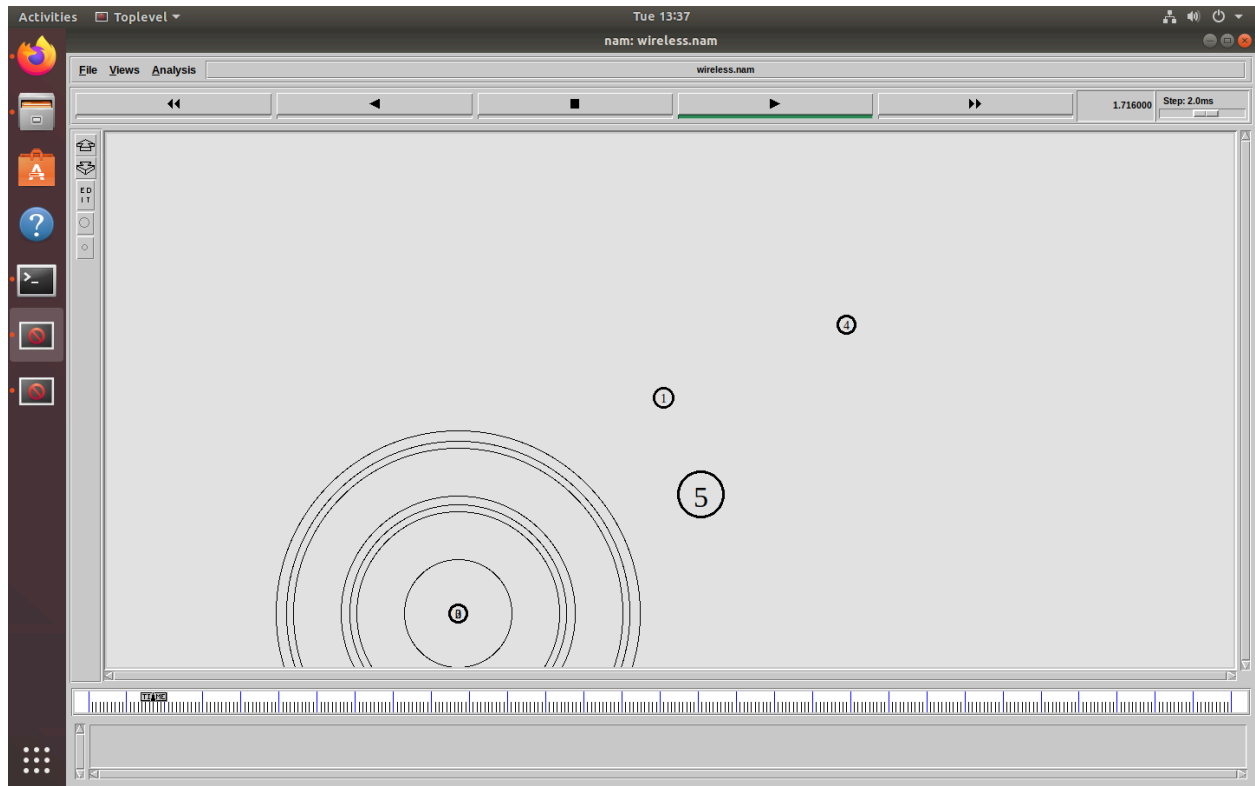
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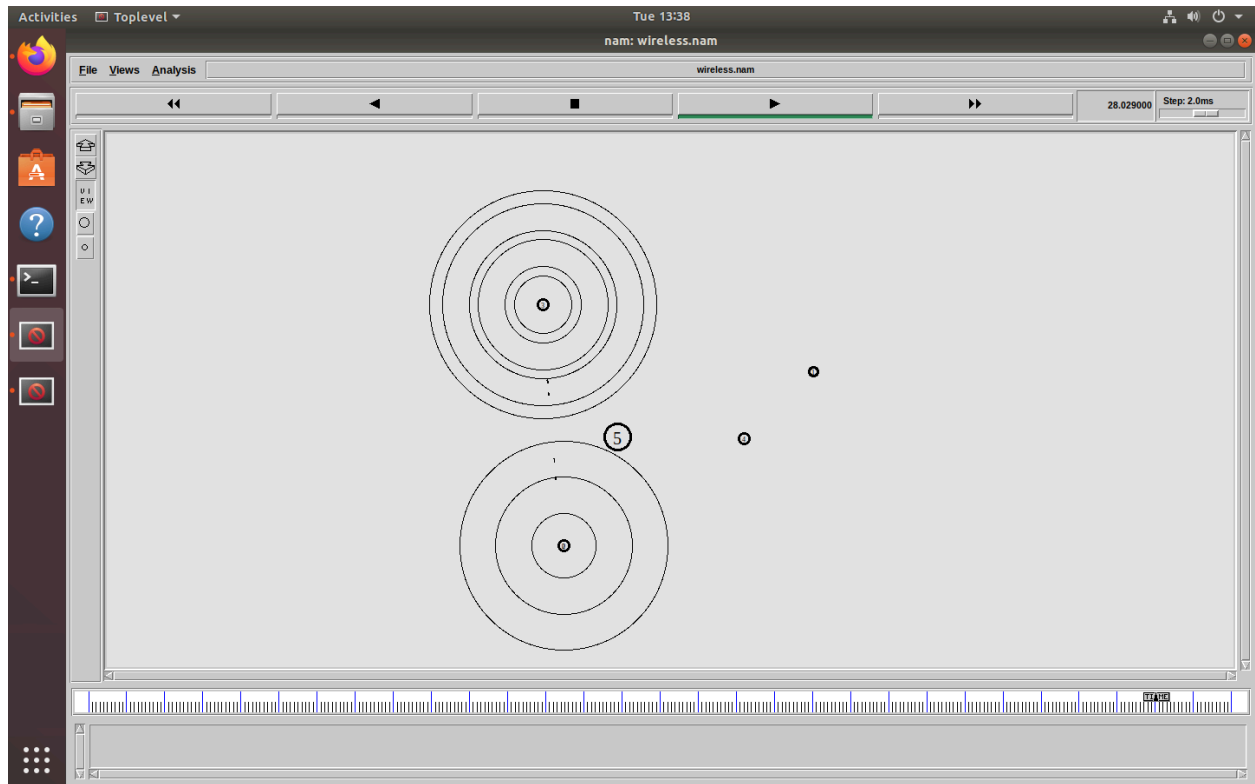
Output: Terminal

```

guest-Surkri@cnlab:~$ gedit wireless_simulation.tcl
guest-Surkri@cnlab:~$ ns wireless_simulation.tcl
When configured, ns found the right version of tclsh in /usr/bin/tclsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the fi
rst tclsh in your path. The wrong version of tclsh may break the test suites. R
econfigure and rebuild ns if this is a problem.
num_nodes is set 6
INITIALIZE THE LIST xListHead
Starting Simulation
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5,  distCST_ = 550.0
SORTING LISTS ...DONE!
guest-Surkri@cnlab:~$ nam wireless.nam
Cannot connect to existing nam instance. Starting a new one...

```





Inference:

1. The simulation demonstrates the operation of a mobile ad-hoc wireless network where nodes can communicate without fixed infrastructure.
2. AODV dynamically establishes routes between source and destination nodes as required.
3. The mobility of nodes changes the network topology during simulation, requiring the routing protocol to discover and maintain routes dynamically.
4. The generated trace file can be analyzed to determine parameters such as packet delivery ratio, throughput, packet loss, and end-to-end delay.

Result: The simulation and analysis of the wired network using the NS2 simulator were successfully completed.